GIS Assignment 3

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Question 1

Summary This Question shows climate change in the USA by average SPEI index across regions in the past 50 years.

```
# Load in shp file
states <- st_read("dataQ1/cb_2018_us_state_20m.shp", quiet = TRUE) |>
  filter(!STUSPS %in% c("AK", "HI", "PR")) |>
  st_transform(crs(spei_us))
```

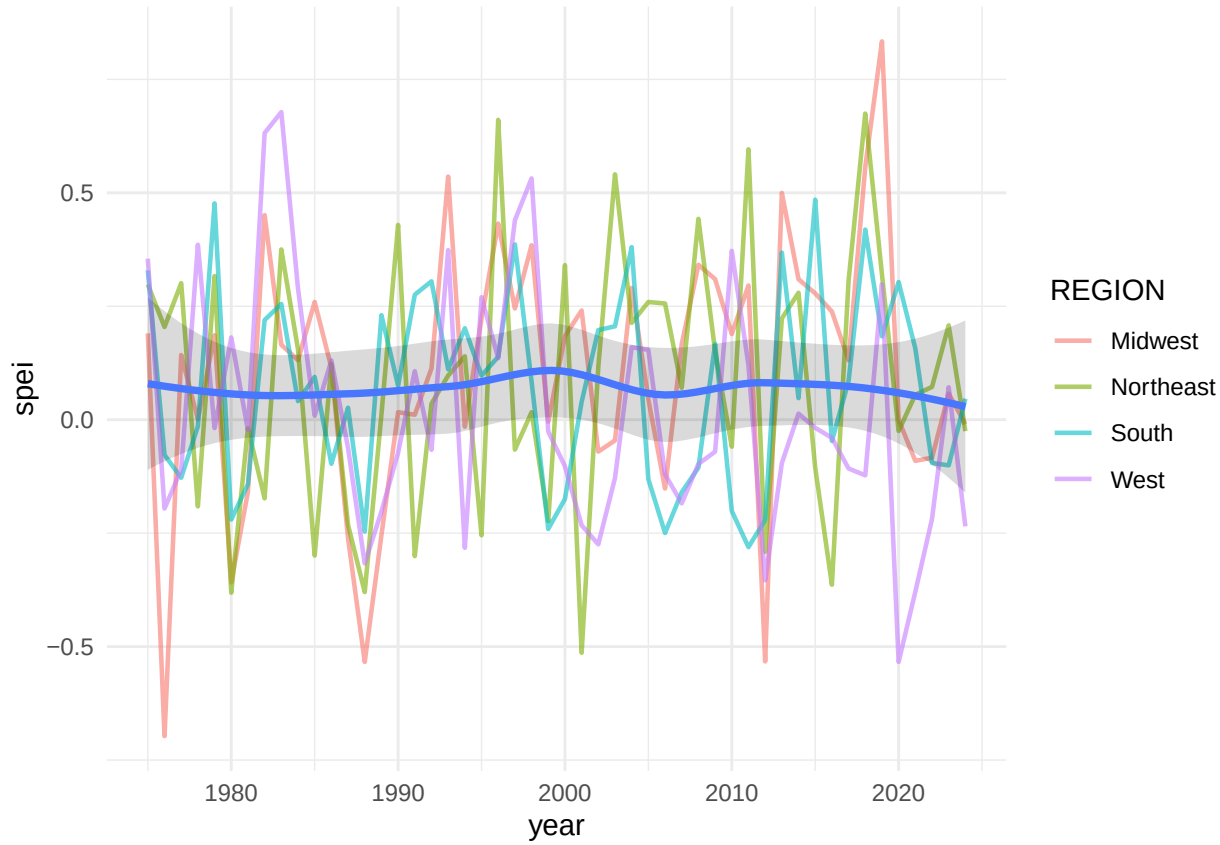
```
region_map <- c(
  # Northeast
  CT="Northeast", ME="Northeast", MA="Northeast", NH="Northeast", RI="Northeast", VT="Northeast",
  NJ="Northeast", NY="Northeast", PA="Northeast",
  # Midwest
  IL="Midwest", IN="Midwest", MI="Midwest", OH="Midwest", WI="Midwest",
  IA="Midwest", KS="Midwest", MN="Midwest", MO="Midwest", NE="Midwest", ND="Midwest", SD="Midwest",
  # South
  DE="South", FL="South", GA="South", MD="South", NC="South", SC="South", VA="South", WV="South", DC="S
  AL="South", KY="South", MS="South", TN="South",
  AR="South", LA="South", OK="South", TX="South",
  # West (continental)
  AZ="West", CO="West", ID="West", MT="West", NV="West", NM="West", UT="West", WY="West",
  CA="West", OR="West", WA="West"
)

# set variable
states$region <- unname(region_map[states$STUSPS])

regions <- states |>
  filter(!is.na(region)) |>
  group_by(region) |>
  summarise(geometry = st_union(geometry), .groups = "drop")

regions_v <- vect(regions)
```

```
# Weighted(area) mean by region
reg_ex <- terra::extract(spei_us, regions_v, fun = mean, weights = TRUE, na.rm = TRUE)

# Build panel -> region x date x SPEI
```

```r
panel <- reg_ex |>
  as.data.frame() |>
  mutate(region = regions$region[ID]) |>
  pivot_longer(
    cols = -c(ID, region),
    names_to = "date",
    values_to = "spei"
  ) |>
  mutate(date = as.Date(date)) |>
  select(region, date, spei) |>
  arrange(region, date)

# Yearly SPEI for each region
panel_year <- panel |>
  mutate(year = year(date)) |>
  group_by(region, year) |>
  summarise(spei = mean(spei, na.rm = TRUE), .groups = "drop")
```

```r
avg_year <- panel_year |>
  group_by(year) |>
  summarise(us_mean = mean(spei, na.rm = TRUE), .groups = "drop")
```

```r
ggplot() +

  # Regional yearly lines (without smooth or CI)
  geom_line(
    data = panel_year,
    aes(x = year, y = spei, color = region, group = region),
    alpha = 0.6,
    linewidth = 0.8
  ) +


  # US average smooth with confidence interval ribbon
  geom_smooth(
    data = avg_year,
    aes(x = year, y = us_mean),
    method = "loess",
    se = TRUE,
    color = "royalblue1",
    fill = "black",
    alpha = 0.15,
    linewidth = 1.3
  ) +

  labs(
    x = "year",
    y = "spei",
    color = "REGION",
  ) +
  theme_minimal()
```



Question 2

Summary

This report solves Assignment 3: identify the 10 most populated places in mainland Spain, build a friction surface (roads cheap, off-road expensive), compute bilateral least-cost distances, and compare isolation for Madrid vs Vigo.

Packages

1. Load Natural Earth data

```
# Country polygon (10m)
countries <- ne_countries(scale = 10, returnclass = 'sf')
spain <- countries %>% filter(admin == 'Spain' | name == 'Spain')

# Populated places and roads (10m)
places <- ne_download(scale = 10, type = 'populated_places', category = 'cultural', returnclass = 'sf')
roads <- ne_download(scale = 10, type = 'roads', category = 'cultural', returnclass = 'sf')

# Ensure valid geometries
places <- st_make_valid(places)
roads <- st_make_valid(roads)
```

```r
# Project to Spain metric CRS (UTM 30N)
crs_proj <- "EPSG:25830"
spain <- st_transform(spain, crs_proj)
places <- st_transform(places, crs_proj)
roads <- st_transform(roads, crs_proj)
```

2. Keep mainland Spain only & select top 10

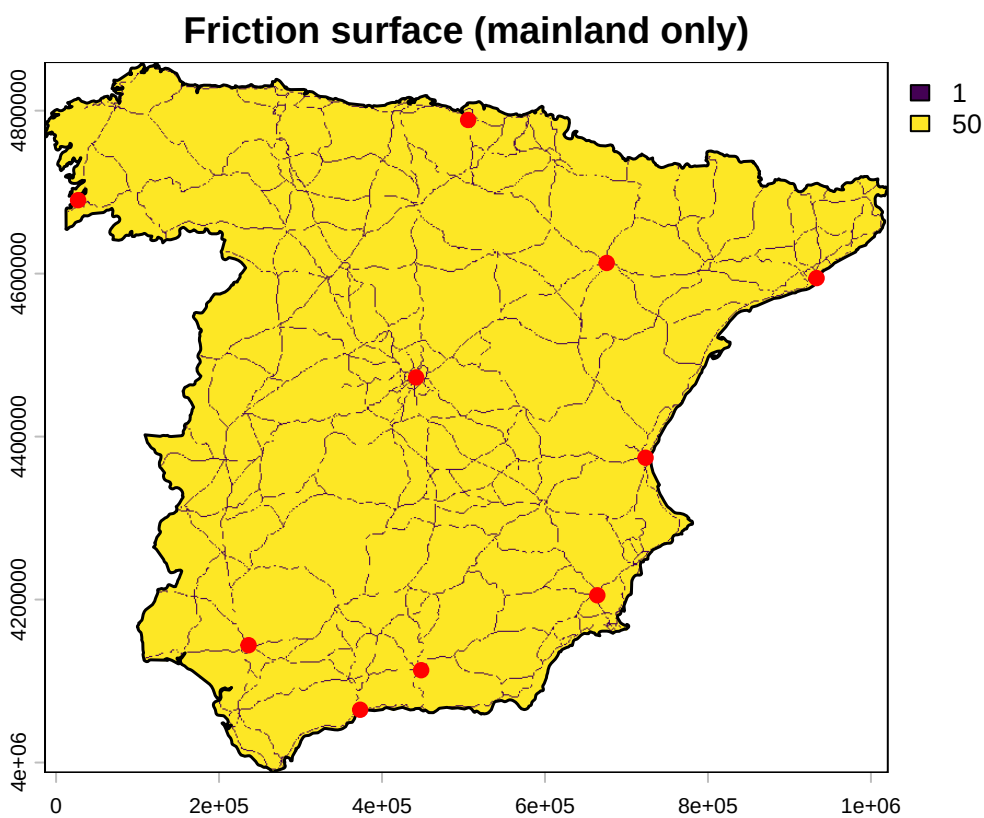
```r
# Keep largest polygon (mainland, exclude islands)
spain_mainland <- spain %>%
  st_cast("MULTIPOLYGON") %>%
  st_cast("POLYGON") %>%
  mutate(area = as.numeric(st_area(.))) %>%
  arrange(desc(area)) %>%
  slice(1) %>%
  dplyr::select(-area)

# Filter places to mainland
places_mainland <- places %>%
  st_join(spain_mainland, left = FALSE)

# Detect population field robustly
cands <- c("POP_MAX","POP2020","POP2015","POP2000","POP_MIN")
pop_field <- cands[cands %in% names(places_mainland)][1]
if (is.na(pop_field)) stop("No population field found.")

places_mainland <- places_mainland %>%
  mutate(pop_est = as.numeric(.data[[pop_field]]))

# Select top 10
top10 <- places_mainland %>%
  arrange(desc(pop_est)) %>%
  slice(1:10)

cat("Top 10 populated places (mainland):\n", paste(top10$NAME, collapse=", "), "\n")
```

```
## Top 10 populated places (mainland):
##  Madrid, Barcelona, Seville, Bilbao, Valencia, Zaragoza, Málaga, Murcia, Granada, Vigo
```

3. Build friction surface from mainland roads

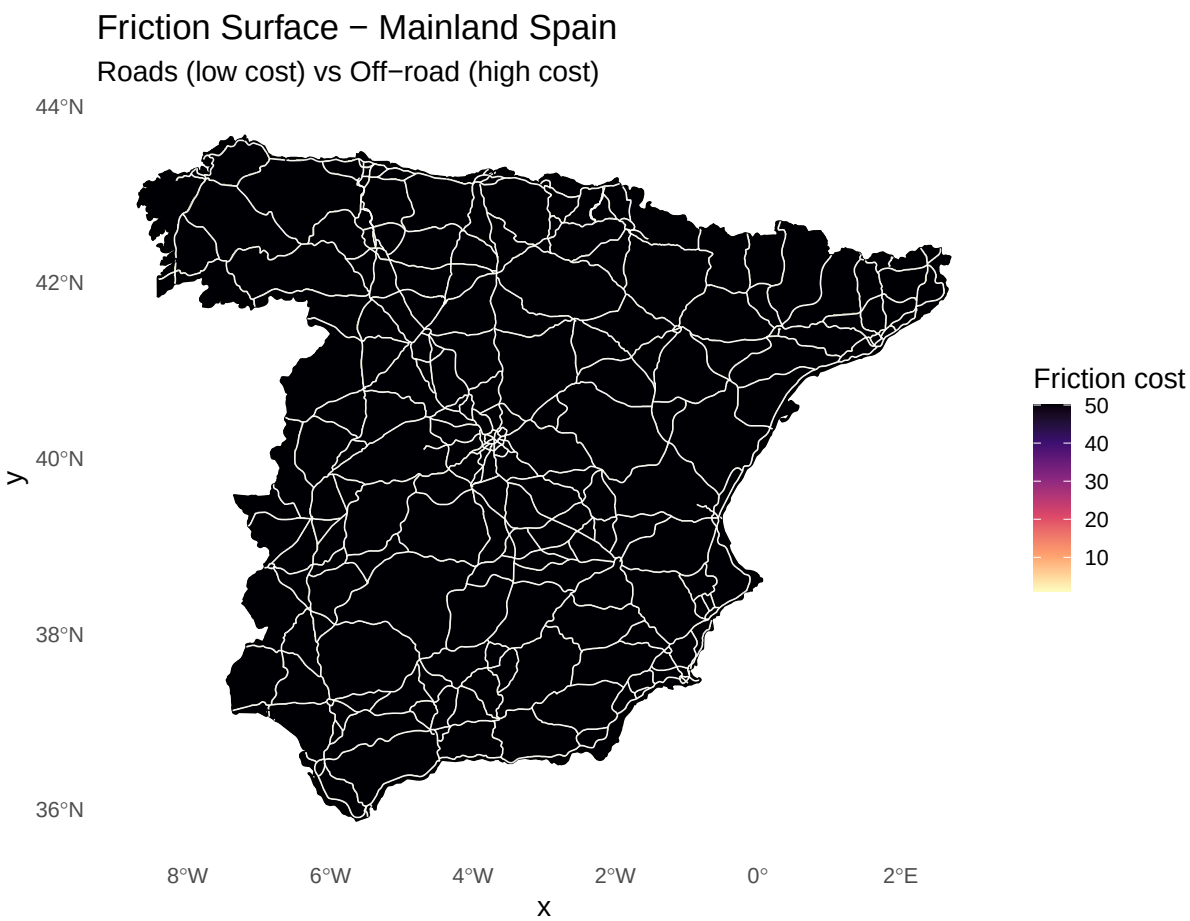
```r
# Clip roads to mainland Spain
roads_mainland <- st_intersection(roads, spain_mainland)

# Create raster template (1 km resolution) over mainland extent
res_m <- 1000
r_template <- rast(ext(vect(spain_mainland)), resolution = res_m, crs = crs_proj)
```

```r
# Initialize with values (NA) before masking
values(r_template) <- NA
r_template <- mask(r_template, vect(spain_mainland))

# Rasterize roads: 1 where exists, NA elsewhere
roads_r <- rasterize(vect(roads_mainland), r_template, field = 1, background = NA)

# Create cost raster: roads cheap (1), off-road expensive (50), sea blocked (NA)
offroad_cost <- 50
road_cost    <- 1

r_cost <- r_template
values(r_cost) <- offroad_cost
values(r_cost)[!is.na(values(roads_r))] <- road_cost

# Final mask to mainland (ensure no sea movement)
r_cost <- mask(r_cost, vect(spain_mainland))

# Plot
plot(r_cost, main = 'Friction surface (mainland only)')
plot(st_geometry(spain_mainland), add = TRUE, border = 'black', lwd = 1.5)
plot(st_geometry(st_centroid(top10)), add = TRUE, col = 'red', pch = 20, cex = 1.5)
```



3b. Friction surface map (modern)

```r
# Convert raster to dataframe for ggplot
r_df <- as.data.frame(r_cost, xy = TRUE, na.rm = TRUE)
colnames(r_df)[3] <- "cost"

# Modern friction surface visualization
ggplot() +
  # Friction raster
  geom_raster(data = r_df,
              aes(x = x, y = y, fill = cost)) +

  # Mainland border
  geom_sf(data = spain_mainland,
          fill = NA,
          color = "black",
          linewidth = 0.4) +

  # Roads overlay (highlighted)
  geom_sf(data = roads_mainland,
          color = "white",
          linewidth = 0.35,
          alpha = 0.9) +

  # Modern color scale (roads bright, off-road darker)
  scale_fill_viridis_c(
    option = "magma",
    direction = -1,
    name = "Friction cost"
  ) +

  coord_sf() +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "right",
    panel.grid = element_blank()
  ) +
  labs(
    title = "Friction Surface - Mainland Spain",
    subtitle = "Roads (low cost) vs Off-road (high cost)"
  )
```



4. Compute 10×10 bilateral least-cost distance matrix

```
# Convert terra SpatRaster to raster::Raster for gdistance
r_cost_raster <- raster::raster(r_cost)

# Build transition object: conductance = 1/cost
tr <- transition(
  x = r_cost_raster,
  transitionFunction = function(x) 1 / mean(x),
  directions = 8
)
tr <- geoCorrection(tr, type = "c")

# Convert top10 to Spatial for gdistance
top10_sp <- as(top10, "Spatial")

# Compute FULL 10x10 bilateral matrix
dist_mat <- costDistance(tr, top10_sp) %>% as.matrix()

# Label rows/columns
city_names <- top10$NAME
rownames(dist_mat) <- city_names
```

```r
colnames(dist_mat) <- city_names

# Display matrix
knitr::kable(dist_mat, caption = "Least-cost distance matrix (10×10, friction units)")
```

Table 1: Least-cost distance matrix (10×10, friction units)

	Madrid	Barcelona	Seville	Bilbao	Valencia	Zaragoza	Málaga	Murcia	Granada	Vigo
Madrid	0.0	603937.2	597545.9	394149.8	384642.1	335014.4	570688.6	435170.6	573580.9	905213.8
Barcelona	603937.2	0.0	1084858.0	613531.6	375317.6	332977.9	1007661.8	606670.7	990792.7	1435535.1
Seville	597545.9	1084858.0	0.0	955492.2	759125.8	927733.3	359257.2	634181.4	486006.1	1310756.6
Bilbao	394149.8	613531.6	955492.2	0.0	634303.4	334369.9	961181.1	823662.6	964073.4	944366.5
Valencia	384642.1	375317.6	759125.8	634303.4	0.0	360579.0	681929.5	283767.8	665060.5	1289855.9
Zaragoza	335014.4	332977.9	927733.3	334369.9	360579.0	0.0	898532.9	593346.8	901425.2	1156373.4
Málaga	570688.6	1007661.8	359257.2	961181.1	681929.5	898532.9	0.0	507540.8	359365.5	1471659.3
Murcia	435170.6	606670.7	634181.4	823662.6	283767.8	593346.8	507540.8	0.0	461695.8	1336727.0
Granada	573580.9	990792.7	486006.1	964073.4	665060.5	901425.2	359365.5	461695.8	0.0	1474551.6
Vigo	905213.8	1435535.1	1310756.6	944366.5	1289855.9	1156373.4	1471659.3	1336727.0	1474551.6	0.0

5. Compare isolation: Madrid vs Vigo

```r
# Extract distance vectors for Madrid and Vigo
madrid_vec <- dist_mat["Madrid", setdiff(city_names, "Madrid")]
vigo_vec   <- dist_mat["Vigo",   setdiff(city_names, "Vigo")]

# Summary statistics
iso_tbl <- data.frame(
  Origin = c("Madrid", "Vigo"),
  Mean = c(mean(madrid_vec), mean(vigo_vec)),
  Median = c(median(madrid_vec), median(vigo_vec)),
  Max = c(max(madrid_vec), max(vigo_vec))
)
rownames(iso_tbl) <- NULL

knitr::kable(iso_tbl, digits = 0, caption = "Isolation summary: bilateral least-cost distances (frictio
```

Table 2: Isolation summary: bilateral least-cost distances (friction units)

Origin	Mean	Median	Max
Madrid	533327	570689	905214
Vigo	1258338	1310757	1474552

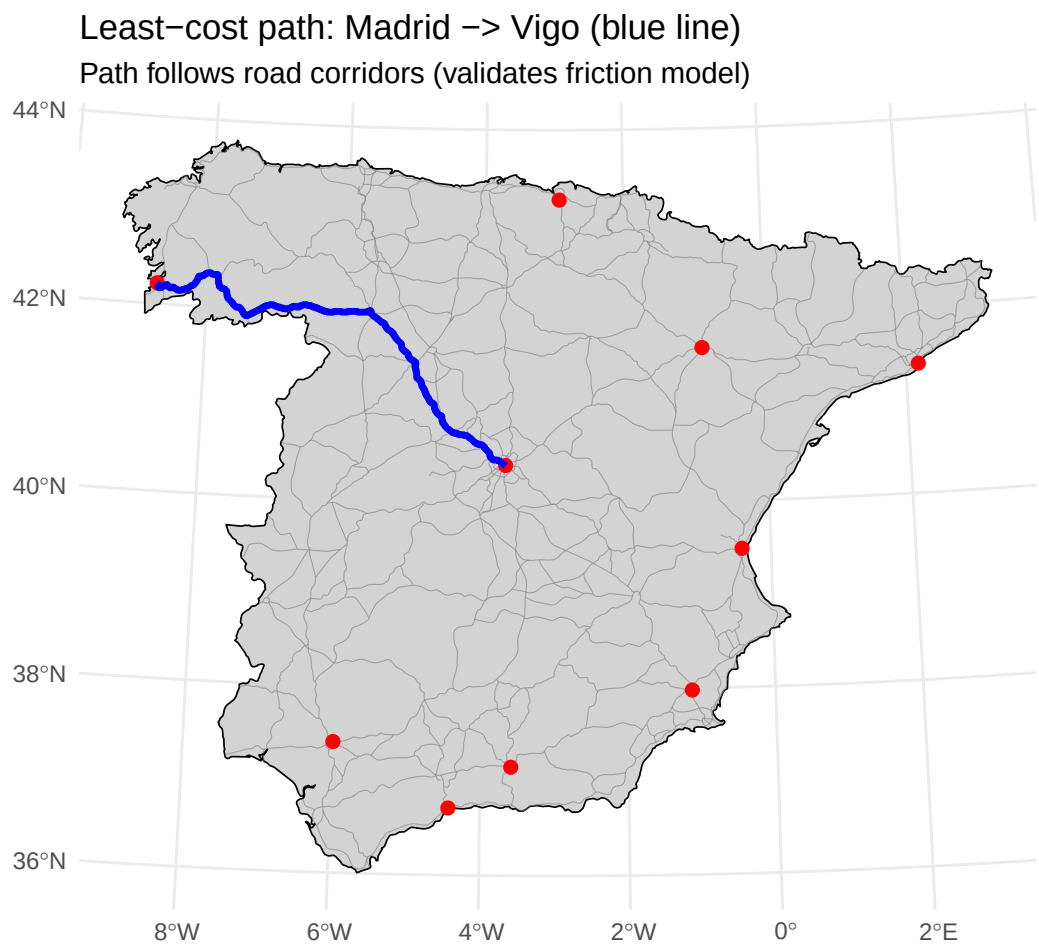
```r
# Determine which is more isolated
madrid_mean <- mean(madrid_vec)
vigo_mean <- mean(vigo_vec)

if (madrid_mean > vigo_mean) {
```

```r
  isolation_result <- paste0("Madrid is more isolated (mean: ", round(madrid_mean, 0),
                             " vs Vigo: ", round(vigo_mean, 0), " friction units).")
} else if (vigo_mean > madrid_mean) {
  isolation_result <- paste0("Vigo is more isolated (mean: ", round(vigo_mean, 0),
                             " vs Madrid: ", round(madrid_mean, 0), " friction units).")
} else {
  isolation_result <- "Madrid and Vigo have equal isolation."
}

cat("**Result:**", isolation_result)
```

```
## **Result:** Vigo is more isolated (mean: 1258338 vs Madrid: 533327 friction units).
```

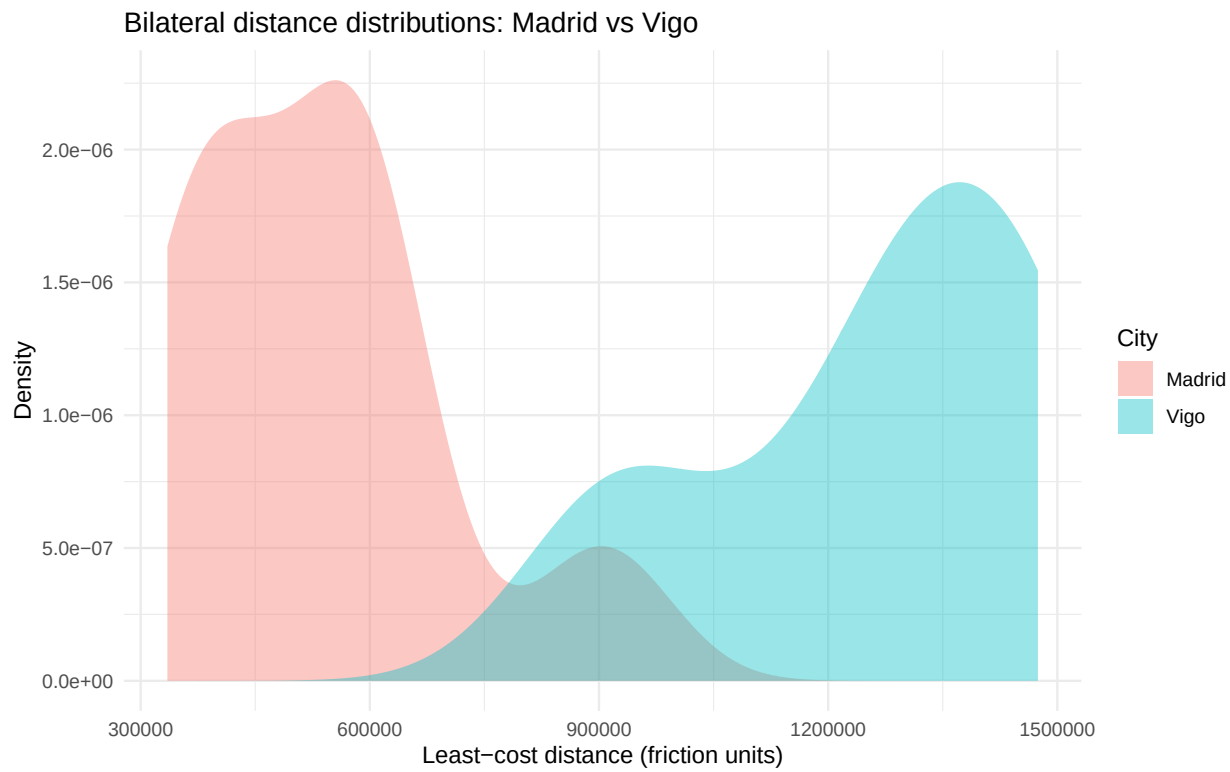
5b. Validation: Madrid → Vigo least-cost path

```r
# Compute shortest (least-cost) path Madrid → Vigo
madrid_cent <- st_centroid(top10 %>% filter(NAME == "Madrid"))
vigo_cent   <- st_centroid(top10 %>% filter(NAME == "Vigo"))

orig_coord <- st_coordinates(madrid_cent)[1, ]
dest_coord <- st_coordinates(vigo_cent)[1, ]

path_mv <- shortestPath(tr, origin = orig_coord, goal = dest_coord, output = "SpatialLines")
path_mv_sf <- st_as_sf(path_mv) %>% st_set_crs(crs_proj)

# Visualize
ggplot() +
  geom_sf(data = spain_mainland, fill = 'lightgrey', color = 'black', linewidth = 0.3) +
  geom_sf(data = roads_mainland, color = 'grey50', linewidth = 0.2, alpha = 0.6) +
  geom_sf(data = st_centroid(top10), color = 'red', size = 2) +
  geom_sf(data = path_mv_sf, color = 'blue', linewidth = 1.2) +
  theme_minimal() +
  labs(title = 'Least-cost path: Madrid → Vigo (blue line)',
       subtitle = 'Path follows road corridors (validates friction model)')
```



5c. Density comparison: Madrid vs Vigo

```
# Prepare data frame for density plot
df_long <- bind_rows(
  data.frame(origin = "Madrid", value = as.numeric(madrid_vec)),
  data.frame(origin = "Vigo",   value = as.numeric(vigo_vec))
)

# Plot overlapping distributions
ggplot(df_long, aes(x = value, fill = origin)) +
  geom_density(alpha = 0.4, color = NA) +
  scale_fill_manual(values = c('Madrid' = '#F8766D', 'Vigo' = '#00BFC4')) +
  theme_minimal() +
  labs(
    x = 'Least-cost distance (friction units)',
    y = 'Density',
    title = 'Bilateral distance distributions: Madrid vs Vigo',
    fill = 'City'
  ) +
  theme(legend.position = 'right')
```



6. Results and conclusions

Assignment objectives achieved:

- **Mainland Spain only**: Islands excluded via polygon filtering
- **Top-10 populated places**: Madrid, Barcelona, Seville, Bilbao, Valencia, Zaragoza, Málaga, Murcia, Granada, Vigo
- **Friction surface**: Roads (cost=1) vs. off-road (cost=50), masked to mainland
- **10×10 bilateral matrix**: Full least-cost distance matrix computed
- **Madrid vs Vigo**: Vigo is more isolated (mean: 1258338 vs Madrid: 533327 friction units).
- **Validation**: Shortest path plotted (confirms road-following behavior)

Summary statistics: - Madrid mean isolation: 5.33327×10^5 friction units - Vigo mean isolation: 1.258338×10^6 friction units - Range of distances: $2.83768 \times 10^5 - 1.474552 \times 10^6$ units

---

Appendix: Save outputs

```
##   Pairwise distance matrix saved to: pairwise_cost_distances_top10.csv
```

```
##   Isolation summary saved to: isolation_summary.csv
```

Notes: - Distances are in **friction units** (not calibrated to meters/km); based on cost surface (road=1, offroad=50). - If data download fails: replace `ne_download()` calls with local shapefiles. - Knit command: `rmarkdown::render('assignment3.Rmd')` in R console.